Relations and Functions

Relation

A **relation** is a set of ordered pairs. Its inputs form the **domain**; its outputs form the **range**.

Function

A **function** assigns *exactly one* output to every input. Different inputs may share the same output.

Function test: ask whether any single input is paired with more than one output. On a graph, this becomes the **vertical line test**.

Decide Whether Each Relation Is a Function

For each relation, give a reason—not only a yes/no answer.

a. $\{(-2, 1), (0, 3), (2, 1), (0, -3)\}$

b. $y^2 = x + 4$

c. A graph passes through $(-3, 2)$, $(0, -1)$, $(2, 4)$, and $(5, 4)$, with no other points.

Find the Domain and Range

Let

$$f(x) = \frac{\sqrt{,5 - 2x,}}{x + 1}.$$

a. State the domain in interval notation.

b. Explain which restriction comes from the square root and which comes from the denominator.

c. Without graphing technology, decide whether 0 belongs to the range. Justify your answer.

Combine Two Functions—Then Track the Domain

Let $f(x) = 2x - 3$ and $g(x) = \sqrt{x+4}$.

a. Find $(f \cdot g)(x)$ and state its domain.

b. Find $\left(\dfrac{f}{g}\right)(x)$ and state its domain.

c. Why are the two domains different even though the same functions are used?

Simplify a Difference Quotient

For $f(x) = x^2 - 3x + 2$, simplify

$$\frac{f(x+h) - f(x)}{h}, \qquad h \neq 0.$$

Your final expression should not have h in the denominator. Clearly show where the common factor of h appears.

What a Graph Can Tell Us

Zeros and intercepts

A real number r is a **zero** of f when $f(r) = 0$. Then $(r, 0)$ is an x-intercept. The y-intercept, if it exists, is found from $f(0)$.

Reading coordinates

$f(a) = b$ means the point (a, b) lies on the graph. Read an input horizontally and its output vertically.

A graph may also reveal domain, range, symmetry, and how often a horizontal or vertical line intersects it. Endpoints and open/closed points matter.

Extract Information from a Formula

Let

$$h(x) = \frac{x^2 - 4}{x - 1}.$$

a. Is $(2, 0)$ on the graph of h? What about $(-2, 0)$?

b. Find the x- and y-intercepts.

c. State the domain in interval notation.

d. Evaluate $h(3)$ and interpret the result as a point on the graph.

Read a Piecewise Graph

Use the graph to answer:

a. $f(-4)$, $f(0)$, and $f(3)$

b. all x- and y-intercepts

c. the domain and range

d. all x such that $f(x) = 2$

e. how many times $x = 1$ intersects the graph